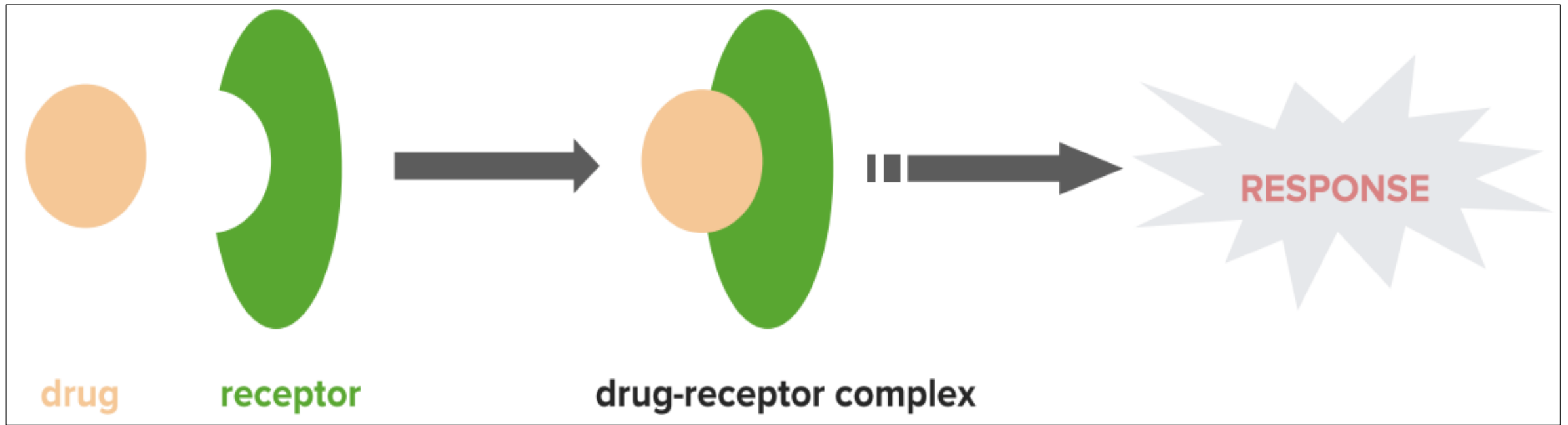




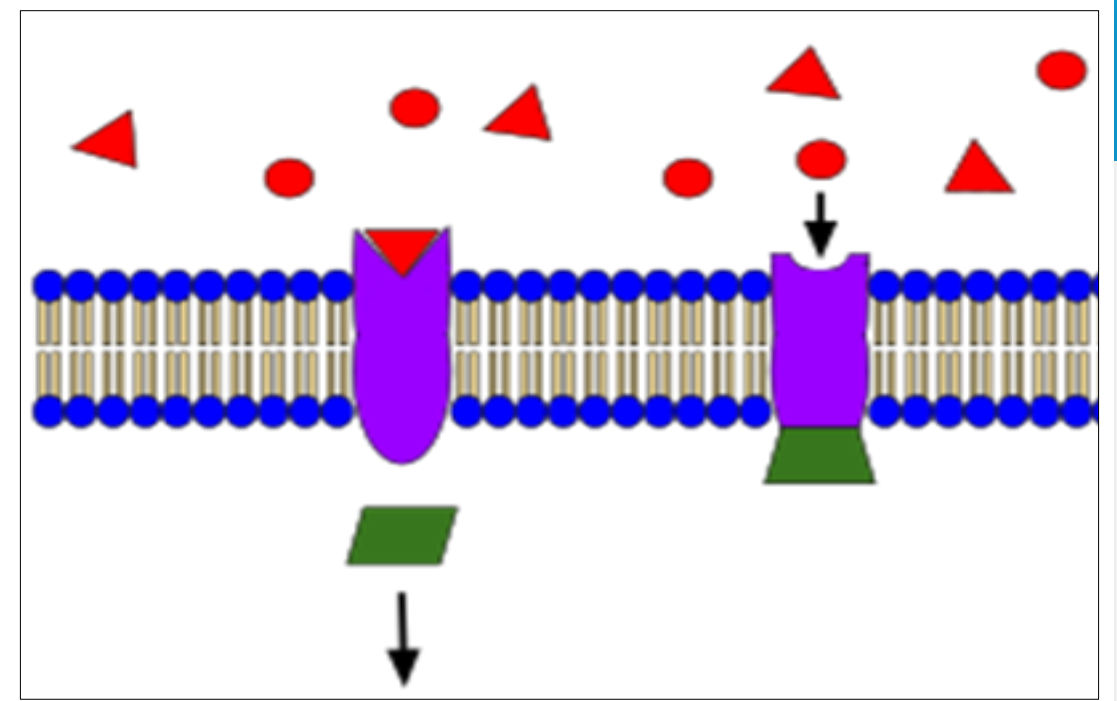
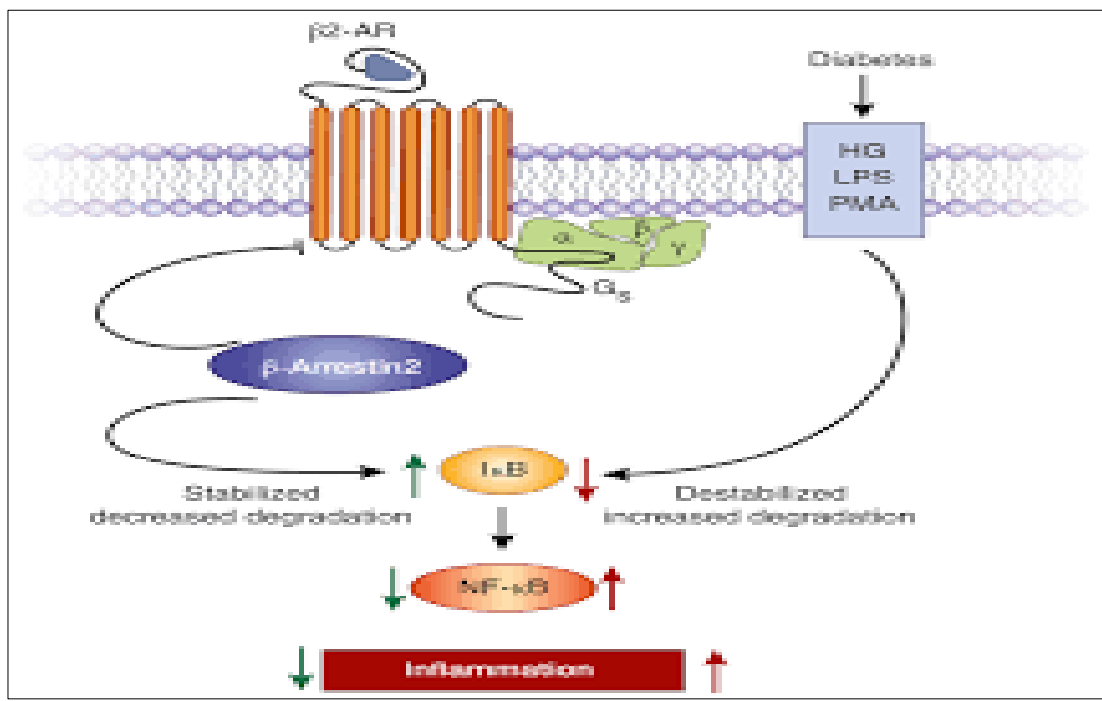
PHARMACOLOGY 1

FACULTY OF NURSING



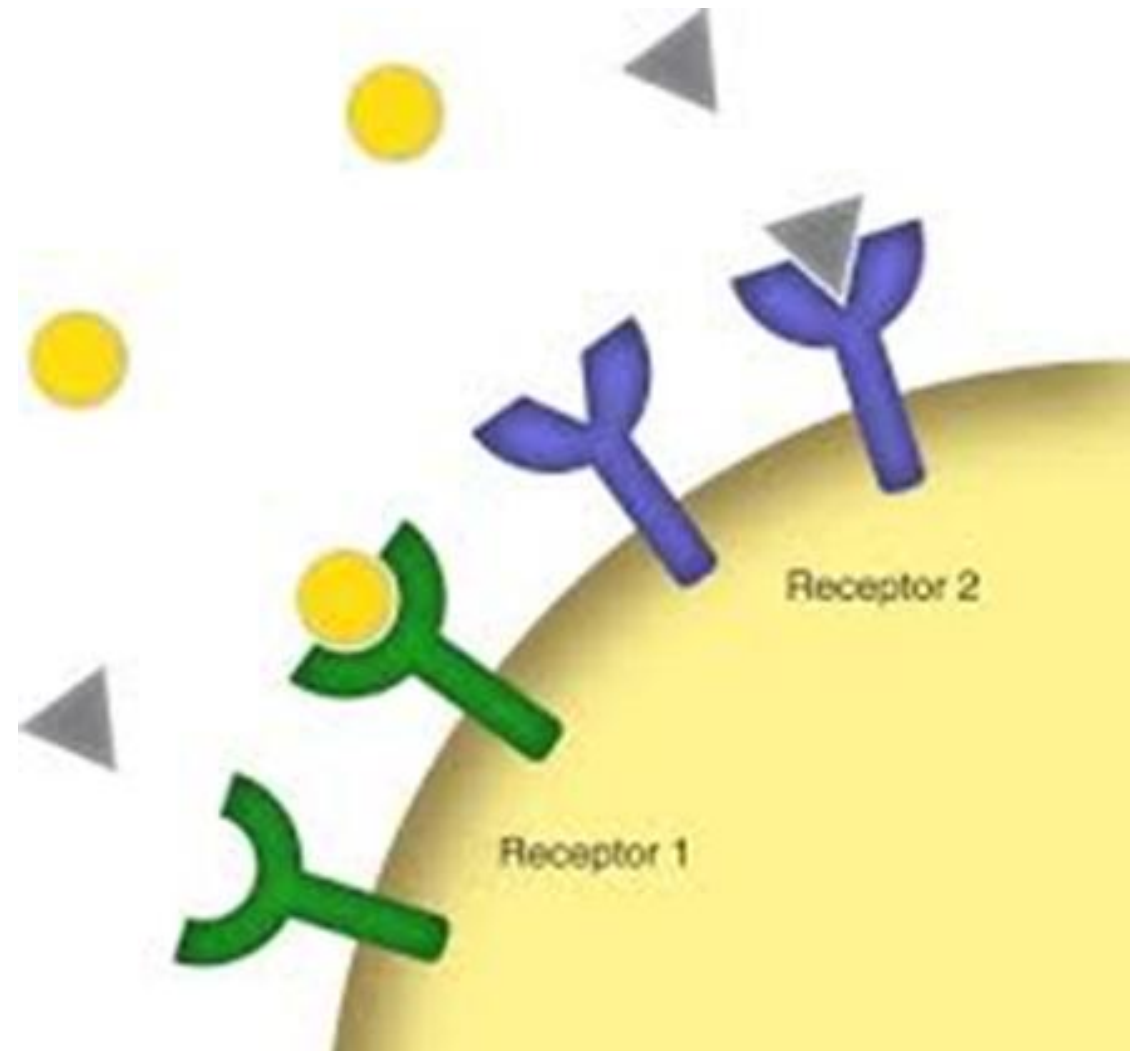
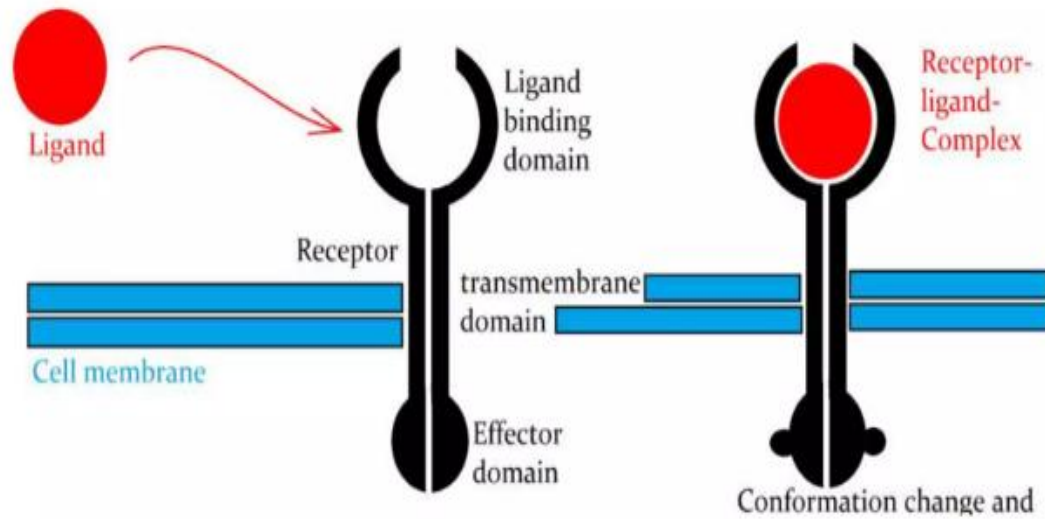
Receptor-mediated Mechanisms

- **Receptors** are **macromolecules** present either on the **cell surface**, **cytoplasm**, or in the **nucleus** with which the drug **binds** and **interacts** to produce **cellular changes**.

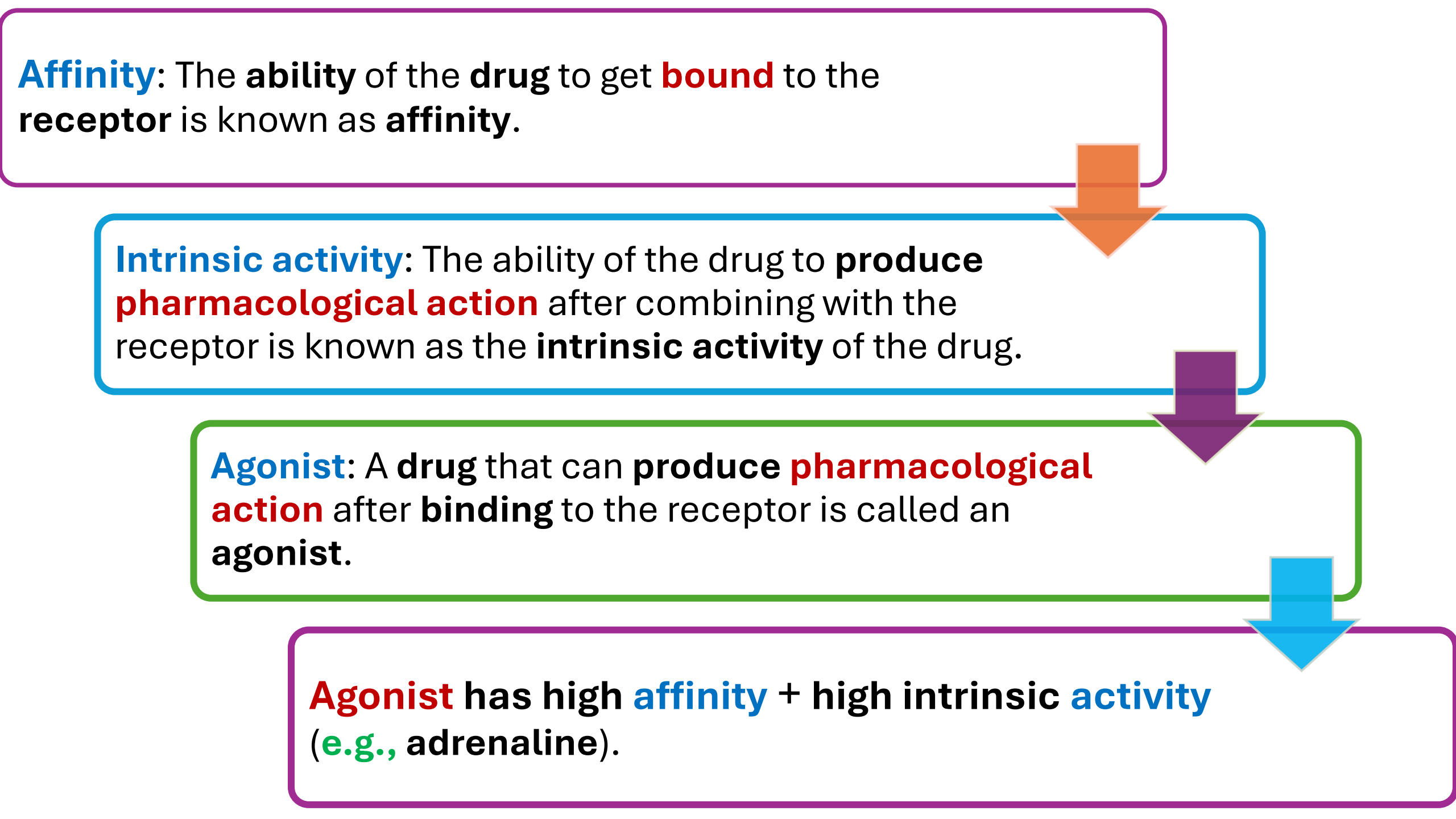


- **For example**, **adrenergic** receptors, and **cholinergic** receptors.

LOCK & KEY MODEL



Affinity: The **ability** of the **drug** to get **bound** to the **receptor** is known as **affinity**.



```
graph TD; A["Affinity: The ability of the drug to get bound to the receptor is known as affinity."] --> B["Intrinsic activity: The ability of the drug to produce pharmacological action after combining with the receptor is known as the intrinsic activity of the drug."]; B --> C["Agonist: A drug that can produce pharmacological action after binding to the receptor is called an agonist."]; C --> D["Agonist has high affinity + high intrinsic activity (e.g., adrenaline)."]
```

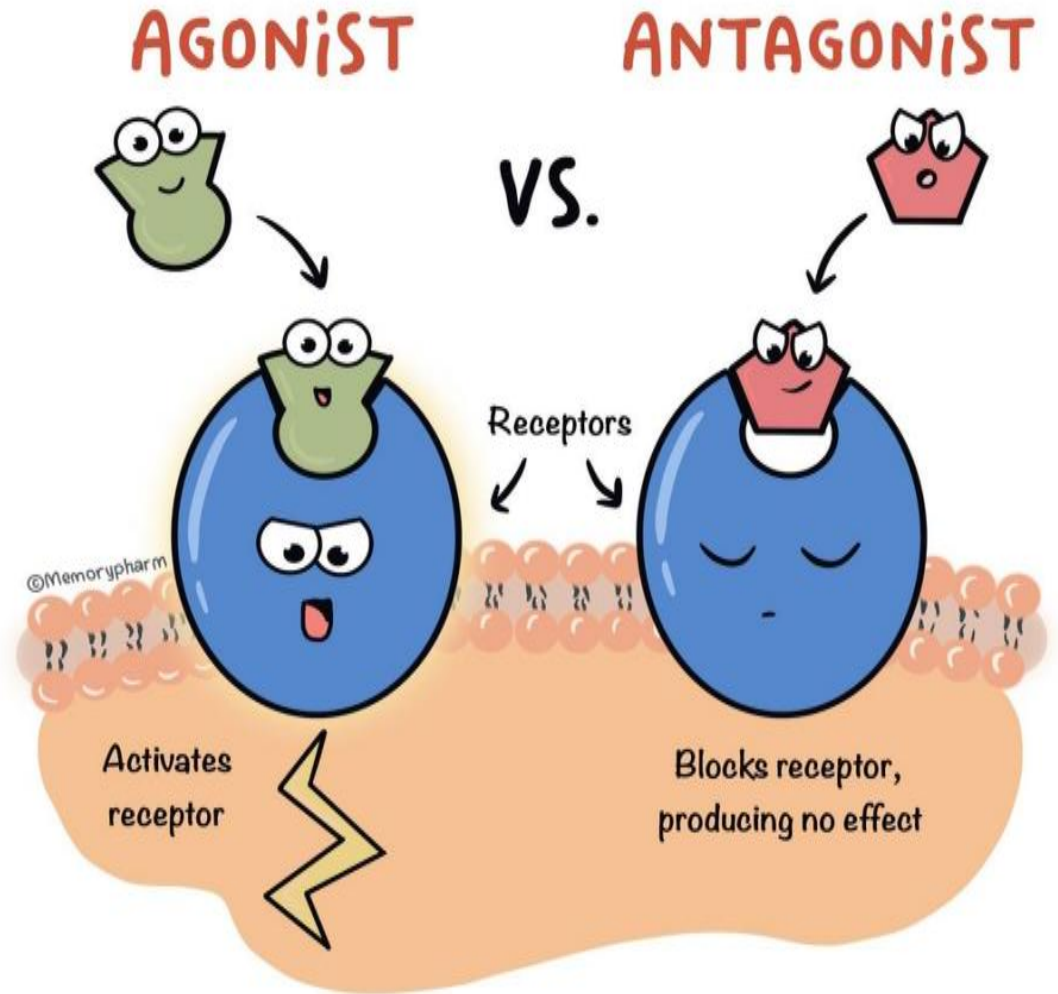
Intrinsic activity: The ability of the drug to **produce pharmacological action** after combining with the receptor is known as the **intrinsic activity** of the drug.

Agonist: A **drug** that can **produce pharmacological action** after **binding** to the receptor is called an **agonist**.

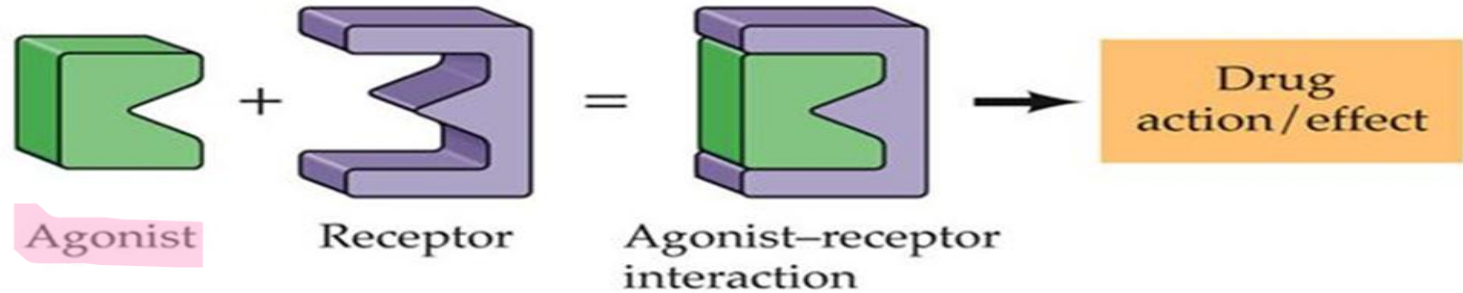
Agonist has high **affinity** + high intrinsic **activity** (e.g., adrenaline).

- **Competitive Antagonist:**

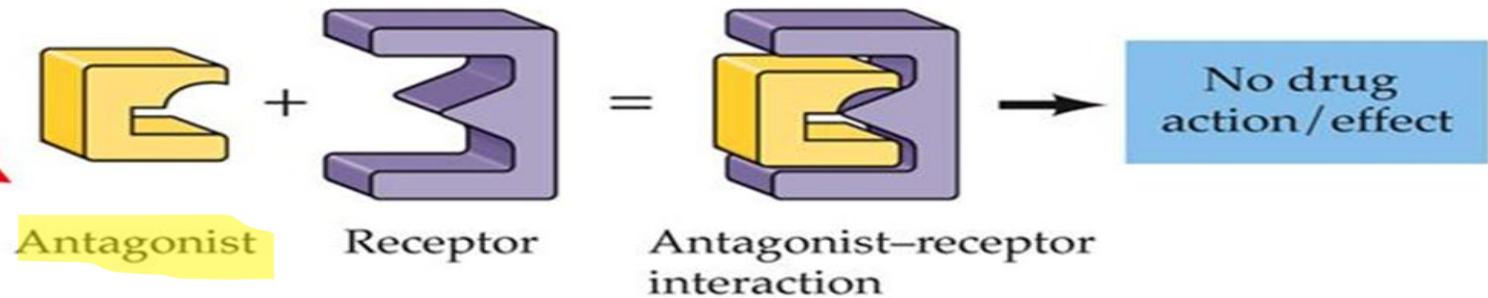
- A drug that **binds** to receptors but is **not** capable of producing **pharmacological action** is called an **antagonist**.
- Antagonist has **high affinity without intrinsic activity** (e.g., atropine).
- It produces **receptor blockade**



Best fit -- highest
affinity

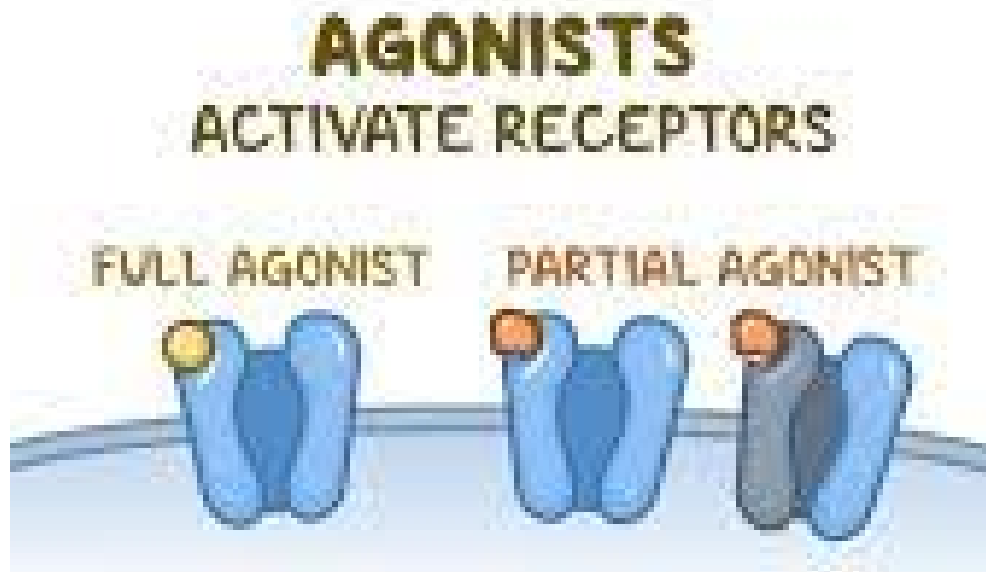


Some fit; no
cellular effect;
block receptor
preventing its
activation by drug
or neurochemical
or hormone



PSYCHOPHARMACOLOGY, Figure 1.11 © 2005 Sinauer Associates, Inc.

The ability of a drug to activate a receptor and generate a cellular response is its efficacy



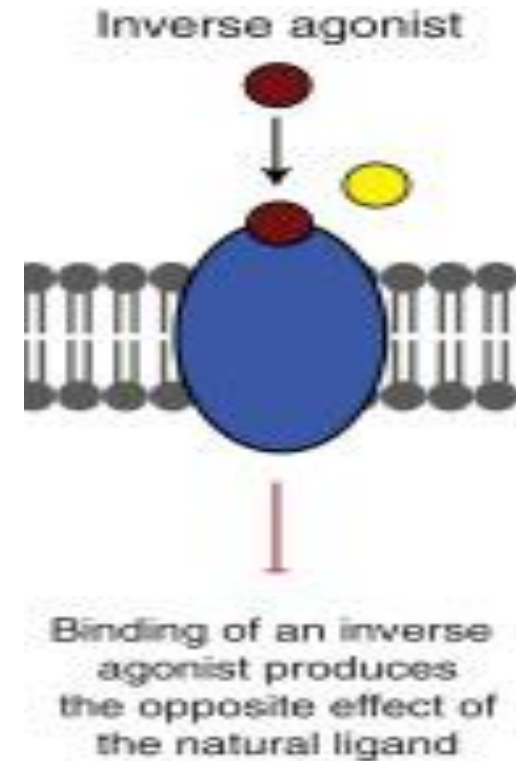
Partial agonist

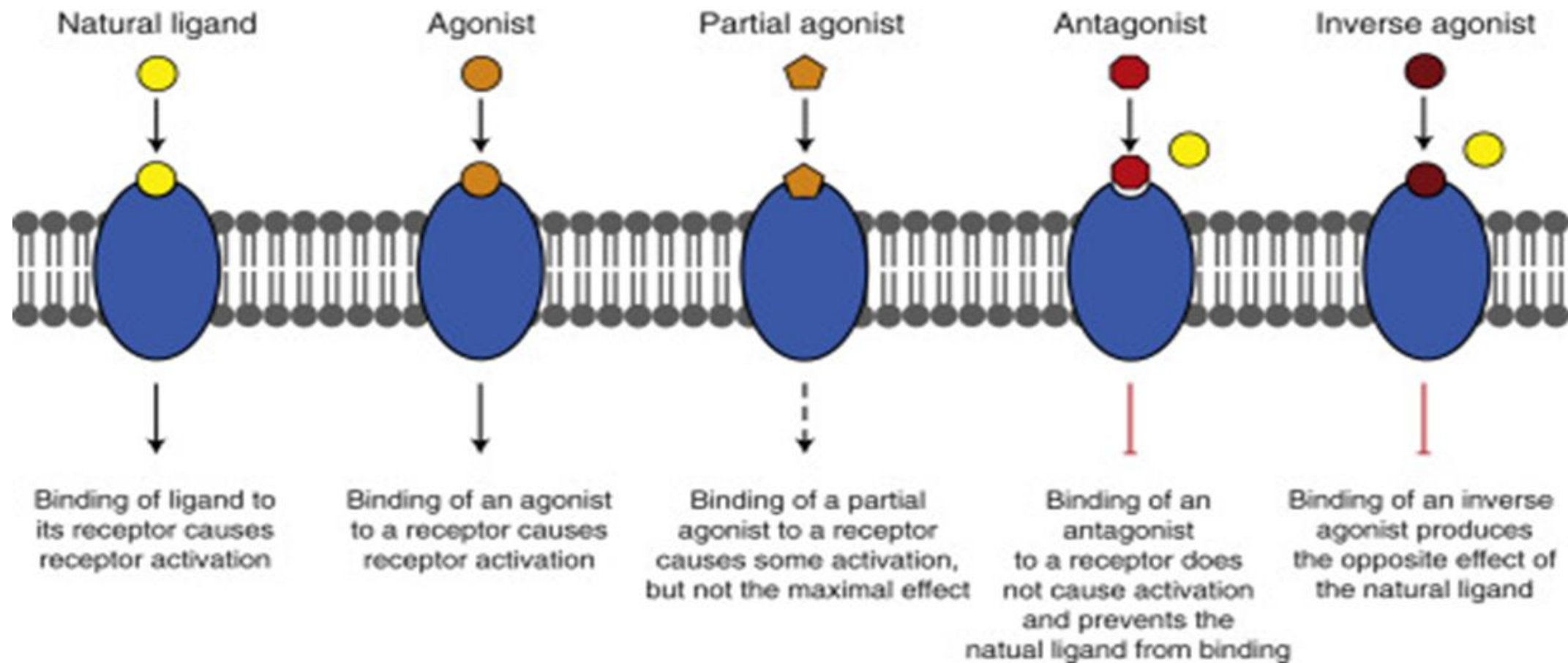
- A drug that binds to the **receptor** but produces an **effect less than** that of an agonist is called a **partial agonist**.

It has **affinity** + **less activity**

Inverse agonist

- It has a full affinity towards the receptor but **produces an effect opposite** to that of an agonist.
- It has an **affinity** and intrinsic **activity** between 0 and -1.





Receptor Families

1. Ligand-gated ion channels (inotropic receptors).

2. G-protein-coupled receptors (metabotropic receptors).

3. Enzymatic receptors.

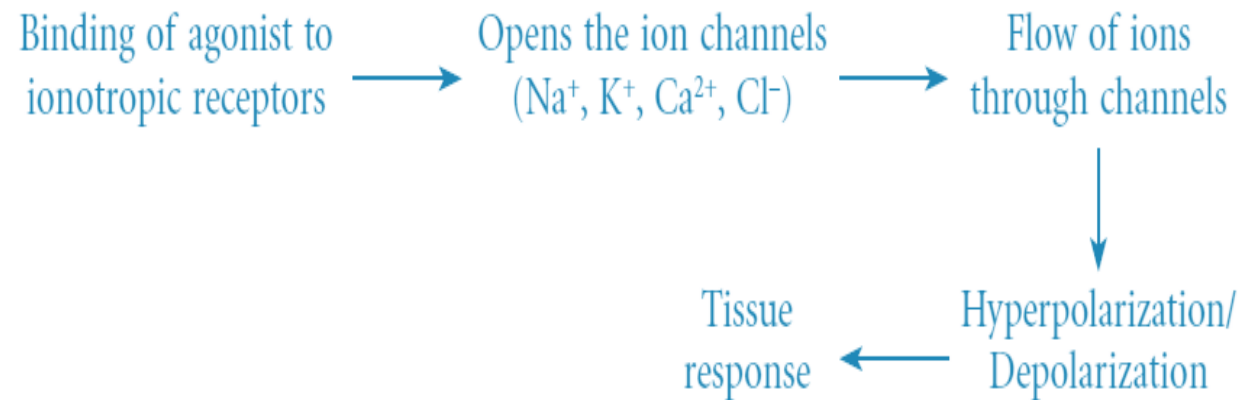
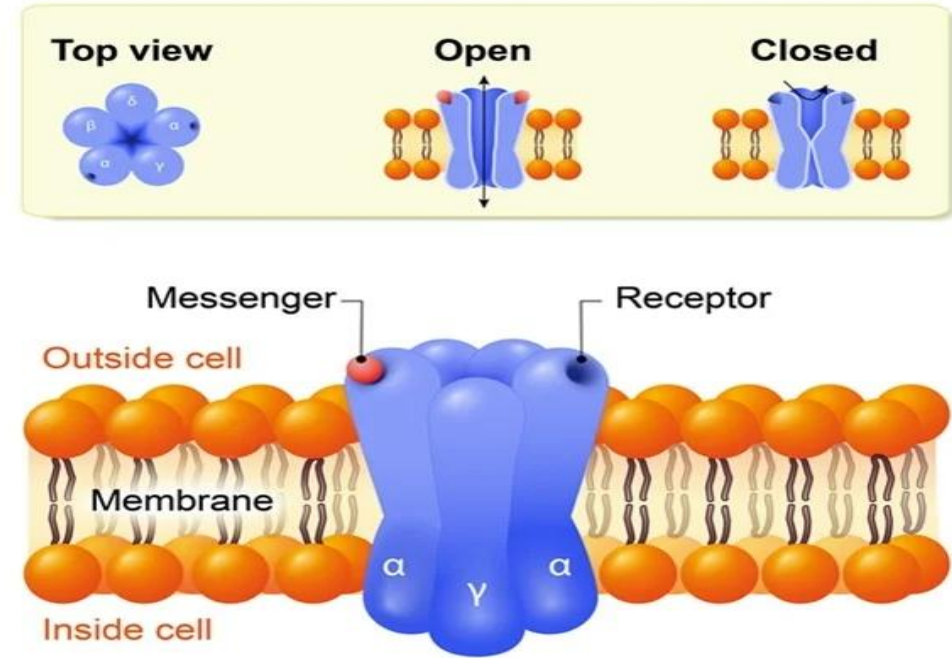
4. Receptor regulating gene expression (transcription factors) or the nuclear receptor.

	Ligand-gated ion Channels	G-protein-coupled Receptors	Enzymatic Receptors	Nuclear Receptors
Location	Membrane	Membrane	Membrane	Intracellular
Effector	Ion channel	Channel or enzyme	Enzyme	Gene transcription
Coupling	Direct	G-proteins (Gs, Gi, Gq, etc.)	Direct	Via DNA
Examples	Nicotinic, GABA _A -receptors	Muscarinic, adrenergic receptors	Insulin, growth factor, cytokine receptors	Steroid, thyroid hormone receptors
Time required for response	Milliseconds	Seconds	Hours	Hours

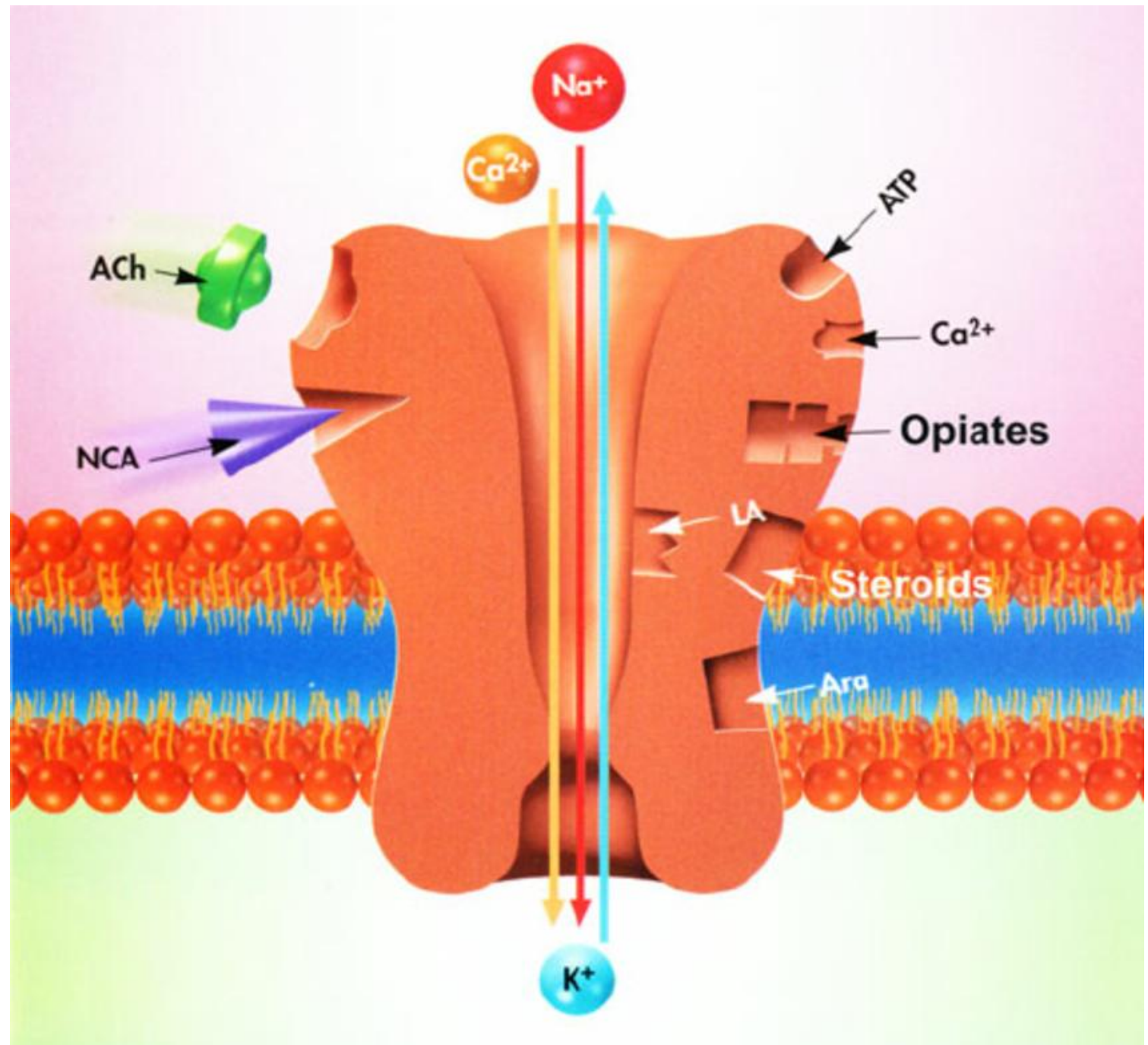
Ligand-gated Ion Channels (Inotropic Receptors)

- **Examples** are
 - ✓ nicotinic (NM) acetylcholine receptors,
 - ✓ GABA receptors in the central nervous system.

ION CHANNEL

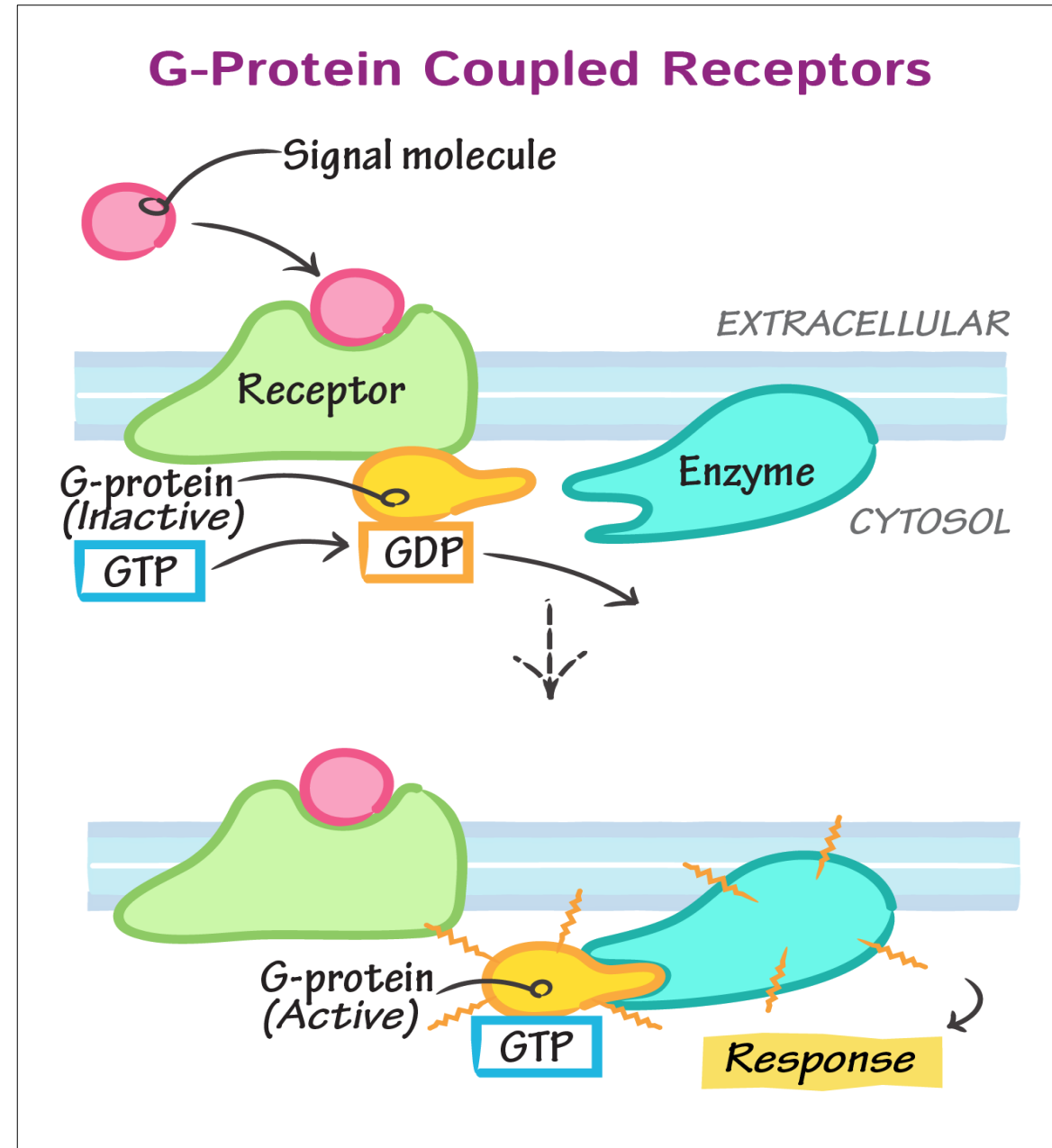


**Nicotinic acetylcholine receptor, a
ligand-gated natural ion channel.**



G-Protein-Coupled Receptors ***(Metabotropic Receptors)***

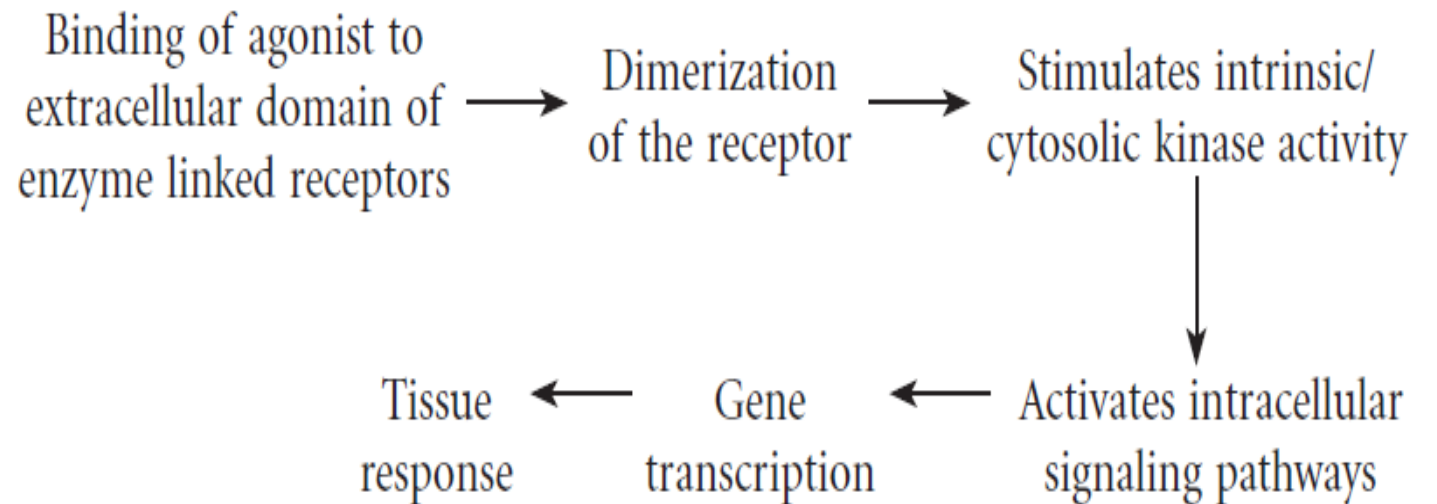
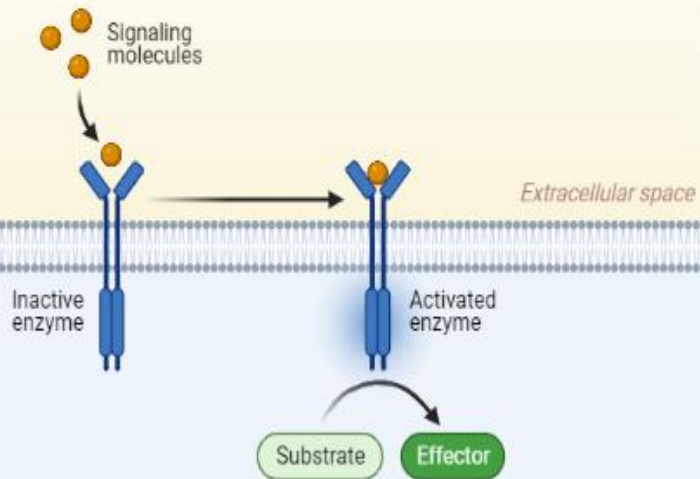
- They are coupled to intracellular effectors through **G-proteins**.
- **G-proteins** are membrane proteins and have three subunits (α, β, γ) with **GDP** bound to α -subunit.



Enzyme-linked Receptors

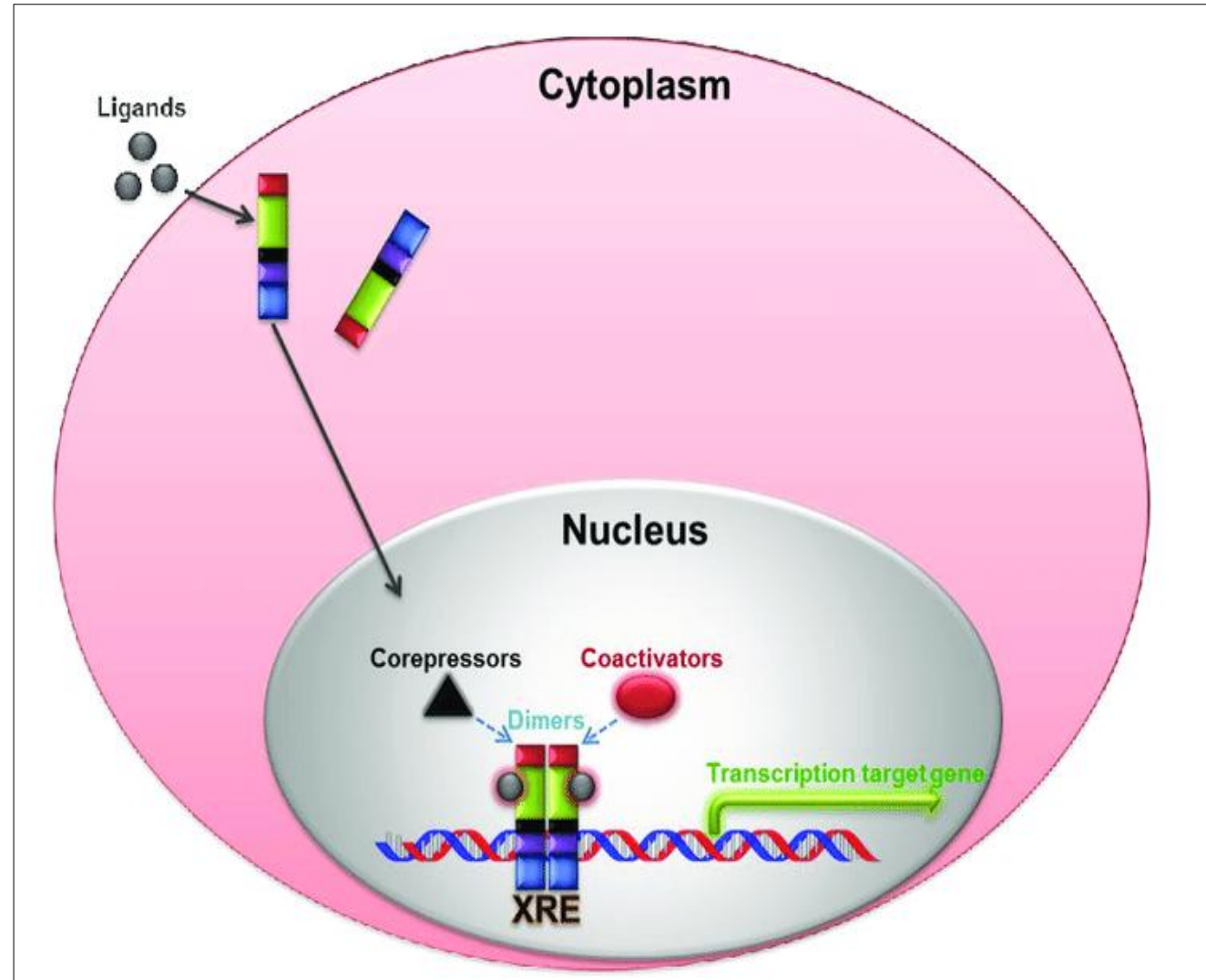
- **For example**, receptors for **insulin** (receptor tyrosine kinase)

Enzyme-Linked Receptors



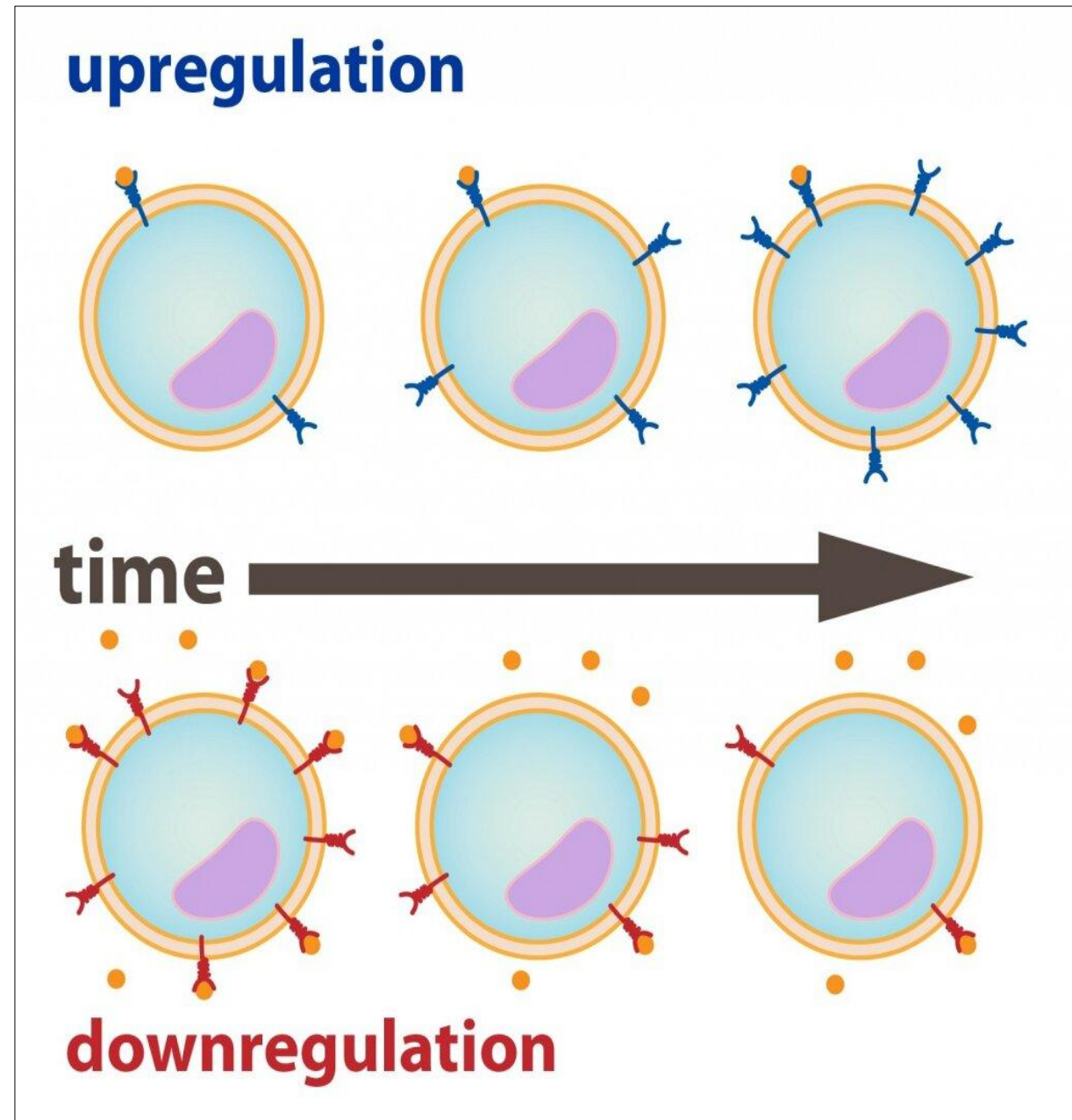
Nuclear Receptors

- **Examples** are receptors for **sex steroids** and **glucocorticoids**.



Regulation of Receptors

- **Receptors** can be regulated by various mechanisms resulting in either their
 - ✓ up-regulation or
 - ✓ downregulation.



Regulation of Receptors

Receptor Down-regulation	Receptor Up-regulation
Prolonged use of agonists ↓ ↓↓ Receptor number and sensitivity ↓ ↓↓ Drug effect For example, chronic use of salbutamol down-regulates the β_2-adrenoceptors, which may be responsible for decreased effect of salbutamol in asthmatics.	Prolonged use of antagonists ↓ ↑↑ Receptor number and sensitivity On sudden stoppage of antagonist ↓ ↑↑ Response to agonist For example, when propranolol is stopped after prolonged use, some patients experience withdrawal symptoms, such as nervousness, anxiety, palpitation, tachycardia, rise in BP, increased incidence of angina or even MI may be precipitated. This is due to up-regulation or supersensitivity of β-adrenoceptors to catecholamines. Therefore, propranolol should not be discontinued abruptly.